

exp counterbalance

General information

This script creates a stimuli counterbalancing list.

The generated sheet can be seen in /exp1 folder, where s_c1_leftd_sheet1 means this is for condition 1(-20ms), left button for t, sheet1 stands for one exposure-test combination (among 12 such combinations)

234 trials were split into three exposure blocks (54 trials each) and six test blocks (12 trials each). Participants take breaks after every 60 trials, which was always during an exposure block. Finally, participants completed an exit survey and an optional demographics survey.

important decision point: VOT distribution

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(tidyr)  
library(ggplot2)  
library(readr)  
library(magrittr)
```

```
##  
## Attaching package: 'magrittr'  
  
## The following object is masked from 'package:tidyr':  
##  
##   extract
```

Test sheet for cond 1 (-20ms)

Trials: 6 Test blocks, 12 each,

Test stimuli: 12 VOT (-5, 5, 15, 25, 30, 35, 40, 45, 50, 55, 65, 70 ms) * 3 pairs = 36 tokens,

Assignment of conditions: Each participant go through each block 1 (3 minimal pair * each pair present 4 times = 12 VOT locations), block 2 (3 pair * 4 times each pair) ...

```
test.vot <- c(-5, 5, 15, 25, 30, 35, 40, 45, 50, 55, 65, 70)
word.list <- c("dill", "dip", "din")

s.test <- crossing(
  VOT = test.vot,
  word = word.list,
  rep = 1:2
) %>%
mutate(
  # grouping realized through match (), every 4 consecutive VOTs share a quartet number, very clever
  quartet = ((match(VOT, test.vot) - 1) %/% 4) + 1,
  # convert word to integer
  wp.idx = match(word, word.list),
  # Latin-square assignmen
  block = ((quartet + wp.idx - 2) %/% 3) + 1 + 3 * (rep - 1)
) %>%
select(VOT, word, block)

# sanity check
s.test %>%
group_by(block) %>%
summarise(
  n.trials = n(),
  n.VOTs = n_distinct(VOT),
  n.dill = sum(word == "dill")
)
```

```
## # A tibble: 6 x 4
##   block n.trials n.VOTs n.dill
##   <dbl>   <int>   <int>   <int>
## 1     1     12     12     4
## 2     2     12     12     4
## 3     3     12     12     4
## 4     4     12     12     4
## 5     5     12     12     4
## 6     6     12     12     4
```

Exposure sheet (-20ms)

Trials: 3 exposure blocks, 54 each (= 24/d/ + 24/t/ + 6 catch trials 2 for each pair)

Test stimuli: 21 token * 3 sampling * 3 minimal pair = 189 -20 ms condition. /d/: m = 5ms, SD = 8.94; /t/: m = 50ms, SD = 16.4 — 24 /d/ + 24 /t/ (Half labeled, half unlabeled, 21 distinct location/token) * 3 sampling -10 ms condition. /d/: m = 15ms, SD = 8.94; /t/: m = 60ms, SD = 16.4 — 24 /d/ + 24 /t/

(Half labeled, half unlabeled, 21 distinct location/token) * 3 sampling +20 ms condition. /d/: m = 45ms, SD = 8.94; /t/: m = 90ms, SD = 16.4 — 24 /d/ + 24 /t/ (Half

labeled, half unlabeled, 21 distinct location/token) * 3 sampling /d/, /t/ screen location counterbalanced across participants

```
exposure.vot <- seq(-20, 125, 5)

# condition 1
m.d.c1 = 5
sd.d.c1 = 8.94
m.t.c1 = 50
sd.t.c1 = 16.4

get.48tokens.from.vot <- function(m.d, sd.d, m.t, sd.t, seed) {
  set.seed(seed)
  weights.d <- dnorm(exposure.vot, mean = m.d, sd = sd.d)
  weights.t <- dnorm(exposure.vot, mean = m.t, sd = sd.t)
  sample.d <- sample(exposure.vot, size = 24, replace = TRUE, prob = weights.d)
  sample.t <- sample(exposure.vot, size = 24, replace = TRUE, prob = weights.t)
  tibble(VOT = c(sample.d, sample.t),
         center = rep(c("d", "t"), each = 24))
}

get.3exposures.for.each.cond <- function(m.d, sd.d, m.t, sd.t,
                                         seed1 = 1, seed2 = 2, seed3 = 3) {
  t1 <- get.48tokens.from.vot(m.d, sd.d, m.t, sd.t, seed1) %>% mutate(block = 1)
  t2 <- get.48tokens.from.vot(m.d, sd.d, m.t, sd.t, seed2) %>% mutate(block = 2)
  t3 <- get.48tokens.from.vot(m.d, sd.d, m.t, sd.t, seed3) %>% mutate(block = 3)
  bind_rows(t1, t2, t3)
}

catch.trials <- tibble(
  block = rep(1:3, each = 6),
  VOT = NA_real_,
  center = rep(c("left", "left", "left", "right", "right", "right"), times = 3),
  condition = "c1",
  label = rep(c(0, 1, 0, 1, 0, 1), times = 3)
) %>%
  mutate(word = center)

# tokens.c1 <- get.3exposures.for.each.cond(m.d.c1, sd.d.c1, m.t.c1, sd.t.c1) %>% mutate(condition = "c1")
# # counterbalance label and word pair
# group_by(block, center) %>%
# mutate(label = sample(rep(c(0, 1), each = 12))) %>%
# group_by(block, center, label) %>%
# mutate(word = sample(rep(c("dill", "dip", "din"), each = 4))) %>%
# ungroup()

assign.label.word.to.cond <- function(m.d, sd.d, m.t, sd.t, cond.label,
                                     seed1 = 1, seed2 = 2, seed3 = 3) {
  get.3exposures.for.each.cond(m.d, sd.d, m.t, sd.t, seed1, seed2, seed3) %>%
    mutate(condition = cond.label) %>%
    group_by(block, center) %>%

```

```

mutate(label = sample(rep(c(0, 1), each = 12))) %>%
group_by(block, center, label) %>%
mutate(word = sample(rep(c("dill", "dip", "din"), each = 4))) %>%
ungroup()
}

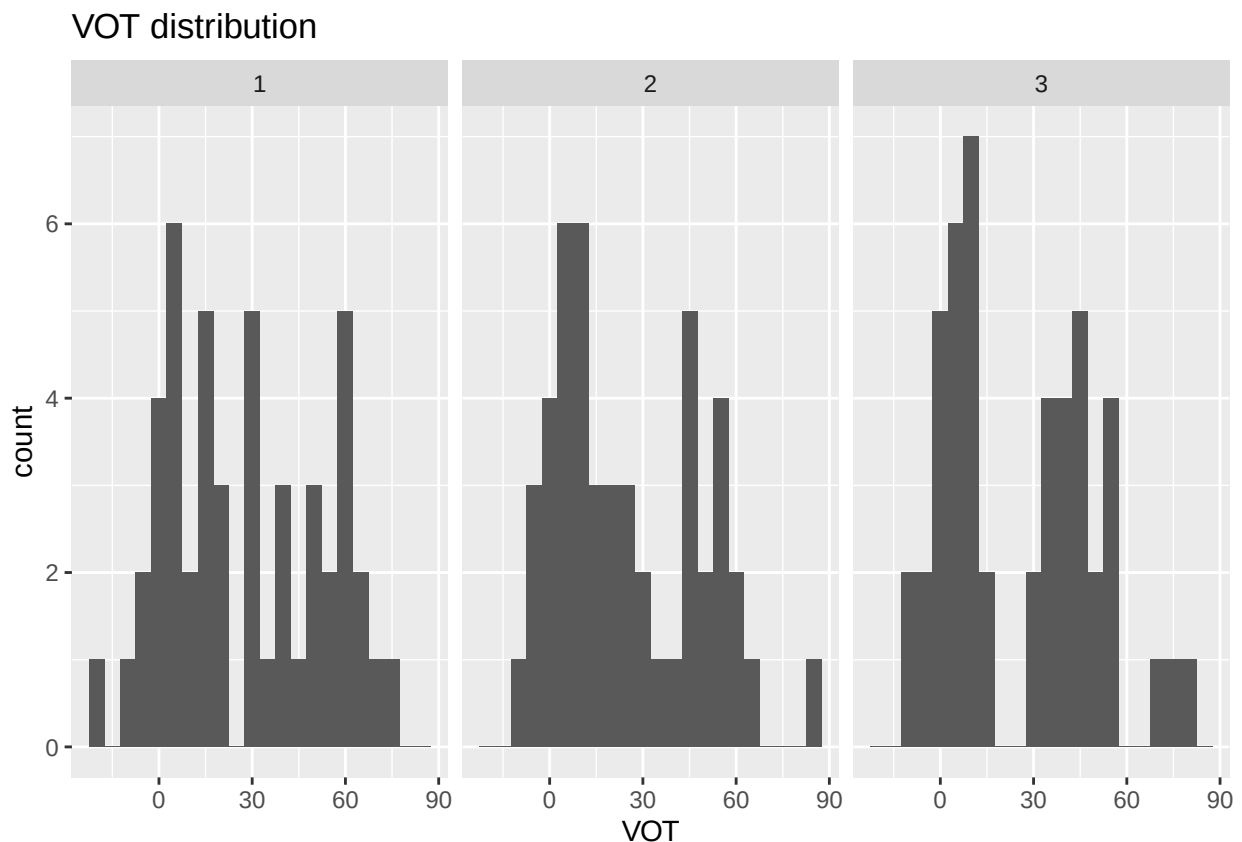
#add catch trials
tokens.c1 <-assign.label.word.to.cond(m.d.c1, sd.d.c1, m.t.c1, sd.t.c1, "c1")
s.c1.exposure <- bind_rows(tokens.c1, catch.trials)
s.c1.exposure %<>% mutate(session = "exposure")
s.test %<>% mutate(session = "test")

s.c1 <- bind_rows(s.c1.exposure, s.test)%>%
  arrange(block, session, VOT)%>%
  mutate(stage = paste0(session, block), randomized_trial = paste0(session, block))

# sanity check the distribution of VOT within s.c1 exposure group
ggplot(s.c1.exposure, aes(x = VOT)) +
  geom_histogram(binwidth = 5) +
  facet_wrap(~block) +
  labs(title = "VOT distribution", x = "VOT", y = "count")

## Warning: Removed 18 rows containing non-finite outside the scale range
## (`stat_bin()`).

```



3 exposure block * 4 test order * 2 left or right = 24 combinations

24 combinations * 3 conditions (-20, -10, +20ms) later = 72 combinations

```
latin.square <- function(n) {
  outer(1:n, 1:n, function(i, j) ((i + j - 2) %% n) + 1)
}
testLS <- latin.square(6)[1:4, ]
exposureLS <- latin.square(3)

# need sanity check
sheet_stage_order <- function(test_idx, exposure_idx) {
  t_perm <- testLS[test_idx, ]
  e_perm <- exposureLS[exposure_idx, ]
  stage_seq <- c(
    paste0("test", t_perm[1]),
    paste0("exposure", e_perm[1]),
    paste0("test", t_perm[2]),
    paste0("exposure", e_perm[2]),
    paste0("test", t_perm[3]),
    paste0("exposure", e_perm[3]),
    paste0("test", t_perm[4]),
    paste0("test", t_perm[5]),
    paste0("test", t_perm[6])
  )
  match(s.c1$stage, stage_seq) # returns 1..9 for each row
}

s.c1 <- s.c1 %>%
  mutate(
    sheet1 = sheet_stage_order(1, 1),
    sheet2 = sheet_stage_order(1, 2),
    sheet3 = sheet_stage_order(1, 3),
    sheet4 = sheet_stage_order(2, 1),
    sheet5 = sheet_stage_order(2, 2),
    sheet6 = sheet_stage_order(2, 3),
    sheet7 = sheet_stage_order(3, 1),
    sheet8 = sheet_stage_order(3, 2),
    sheet9 = sheet_stage_order(3, 3),
    sheet10 = sheet_stage_order(4, 1),
    sheet11 = sheet_stage_order(4, 2),
    sheet12 = sheet_stage_order(4, 3)
  )

cols <- c("VOT", "center", "block", "condition", "label", "word", "stage", "session", "randomized_trial")

##### left for d, output 12csv
s.c1.leftd <- s.c1 %>%
  mutate(left = "d", right = "t")
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c1_leftd_sheet", i, ".csv")
  s.c1.leftd %>%
    arrange(.data[[sheet_col]]) %>%
```

```

    select(all_of(cols)) %>%
    write_csv(filename)
}

##### left for t, output 12csv
s.c1.leftt <- s.c1 %>% #change here
  mutate(left = "t", right = "d") #change here
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c1_leftt_sheet", i, ".csv")
  s.c1.leftt %>%
    arrange(.data[[sheet_col]]) %>%
    select(all_of(cols)) %>%
    write_csv(filename)
}

```

replicate the proces for c2 (-10ms) and c3 condition (20ms)

```

# condition 2 and 3
m.d.c2 = m.d.c1 + 10
m.t.c2 = m.t.c1 + 10

tokens.c2 <- assign.label.word.to.cond(m.d.c2, sd.d.c1, m.t.c2, m.t.c1, "c2")
s.c2.exposure <- bind_rows(tokens.c2, catch.trials)
s.c2.exposure %<>% mutate(session = "exposure")
s.c2 <- bind_rows(s.c2.exposure, s.test)%>%
  arrange(block, session, VOT)%>%
  mutate(stage = paste0(session, block), randomized_trial = paste0(session, block))

s.c2 <- s.c2 %>%
  mutate(
    sheet1 = sheet_stage_order(1, 1),
    sheet2 = sheet_stage_order(1, 2),
    sheet3 = sheet_stage_order(1, 3),
    sheet4 = sheet_stage_order(2, 1),
    sheet5 = sheet_stage_order(2, 2),
    sheet6 = sheet_stage_order(2, 3),
    sheet7 = sheet_stage_order(3, 1),
    sheet8 = sheet_stage_order(3, 2),
    sheet9 = sheet_stage_order(3, 3),
    sheet10 = sheet_stage_order(4, 1),
    sheet11 = sheet_stage_order(4, 2),
    sheet12 = sheet_stage_order(4, 3)
  )

##### left for d, output 12csv
s.c2.leftd <- s.c2 %>%
  mutate(left = "d", right = "t")
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c2_leftd_sheet", i, ".csv")
}

```

```

s.c2.leftd %>%
  arrange(.data[[sheet_col]]) %>%
  select(all_of(cols)) %>%
  write_csv(filename)
}

##### left for t, output 12csv
s.c2.leftt <- s.c2 %>% #change here
  mutate(left = "t", right = "d") #change here
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c2_leftt_sheet", i, ".csv")
  s.c2.leftt %>%
    arrange(.data[[sheet_col]]) %>%
    select(all_of(cols)) %>%
    write_csv(filename)
}

```

```

# condition 2 and 3
m.d.c3 = m.d.c1 + 40
m.t.c3 = m.t.c1 + 40

tokens.c3 <- assign.label.word.to.cond(m.d.c3, sd.d.c1, m.t.c3, m.t.c2, "c2")
s.c3.exposure <- bind_rows(tokens.c3, catch.trials)
s.c3.exposure %<>% mutate(session = "exposure")
s.c3 <- bind_rows(s.c3.exposure, s.test)%>%
  arrange(block, session, VOT)%>%
  mutate(stage = paste0(session, block), randomized_trial = paste0(session, block))

s.c3 <- s.c3 %>%
  mutate(
    sheet1 = sheet_stage_order(1, 1),
    sheet2 = sheet_stage_order(1, 2),
    sheet3 = sheet_stage_order(1, 3),
    sheet4 = sheet_stage_order(2, 1),
    sheet5 = sheet_stage_order(2, 2),
    sheet6 = sheet_stage_order(2, 3),
    sheet7 = sheet_stage_order(3, 1),
    sheet8 = sheet_stage_order(3, 2),
    sheet9 = sheet_stage_order(3, 3),
    sheet10 = sheet_stage_order(4, 1),
    sheet11 = sheet_stage_order(4, 2),
    sheet12 = sheet_stage_order(4, 3)
  )

##### left for d, output 12csv
s.c3.leftd <- s.c3 %>%
  mutate(left = "d", right = "t")
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c2_leftd_sheet", i, ".csv")
  s.c3.leftd %>%
    arrange(.data[[sheet_col]]) %>%

```

```

    select(all_of(cols)) %>%
    write_csv(filename)
}

##### left for t, output 12csv
s.c3.leftt <- s.c3 %>% #change here
  mutate(left = "t", right = "d") #change here
for (i in 1:12) {
  sheet_col <- paste0("sheet", i)
  filename <- paste0("/Users/nwang40/Documents/Gitrepo/Spanish-DL/exp1/s_c3_leftt_sheet", i, ".csv")
  s.c1.leftt %>%
    arrange(.data[[sheet_col]]) %>%
    select(all_of(cols)) %>%
    write_csv(filename)
}

```